[Paper 476, continued: *On the Determination of the Orbit of a Planet from Three Observations*, Articles 64–97.]

[p. 429]

we obtain the six coordinates of the three rays respectively

$$\begin{aligned}(a_1,\, b_1,\, c_1,\, f_1,\, g_1,\, h_1) &= (\,0,\ \sqrt{3},\ -1,\ \ \ 0,\ 1,\ \sqrt{3}),\\ (a_2,\, b_2,\, c_2,\, f_2,\, g_2,\, h_2) &= (\,3,\ \sqrt{3},\ \ 2,\ -\sqrt{3},\ 1,\ -2\sqrt{3}),\\ (a_3,\, b_3,\, c_3,\, f_3,\, g_3,\, h_3) &= (-3,\ \sqrt{3},\ \ 2,\ -\sqrt{3},\ 1,\ -2\sqrt{3}),\end{aligned}$$

whence the intersections with the orbit-plane are given by

$$\begin{aligned}x_1' : y_1' : 1 &= \beta'\sqrt{3} - \gamma' \ :\ -\beta'\sqrt{3} + \gamma \ :\ \beta'' + \gamma''\sqrt{3},\\ x_2' : y_2' : 1 &= 3\alpha' + \beta'\sqrt{3} + 2\gamma' \ :\ -3\alpha - \beta\sqrt{3} - 2\gamma \ :\ \alpha''\sqrt{3} + \beta'' - 2\sqrt{3}\gamma'',\\ x_3' : y_3' : 1 &= -3\alpha' + \beta'\sqrt{3} + 2\gamma' \ :\ 3\alpha - \beta\sqrt{3} - 2\gamma \ :\ -\alpha''\sqrt{3} + \beta'' - 2\sqrt{3}\gamma'',\end{aligned}$$

where if (as before) the position of the orbit-plane be determined by means of the longitude b and colatitude c of the orbit-pole, we have

$$\begin{aligned}\alpha\ ,\ \beta\ ,\ \gamma &= \sin b\ ,\ -\cos b\ ,\ 0\ ,\\ \alpha',\ \beta',\ \gamma' &= \cos b\cos c,\ \sin b\cos c,\ -\sin c,\\ \alpha'',\ \beta'',\ \gamma'' &= \cos b\sin c,\ \sin b\sin c,\ \cos c,\end{aligned}$$

and the passage from the coordinates x', y', to x, y, is given by

$$\begin{aligned}x' &= x\sin b - y\cos b,\\ y' &= +x\cos b + y\sin b,\end{aligned}$$

or conversely

$$\begin{aligned}x &= x'\sin b + y'\cos b,\\ y &= -x'\cos b + y'\sin b.\end{aligned}$$

65. To develope the results, I consider the orbit-pole as passing through certain series of positions. The locus may be a meridian circle: by reason of the symmetry of the system, the results are not altered by a change of 120° in the longitude of the meridian; so that, by considering the two meridians 0°–180° and 90°–270°, we, in fact, consider twelve half meridians at the intervals of 30°. An illustration is afforded by Plate I.; the orbit-pole describes successively the meridians 0°, 30°, 60°, 90°, and the line 1, by its intersection with the orbit-plane, traces out on this plane a series of hyperbolas shown in the figure; the hyperbola for the meridian 90° is a right line, but (except for the position where the orbit-plane passes through the line 1) the locus is a determinate point on this line. Pianogram No. 1 (Plate II.) refers to the meridian 90°–270°, and Pianogram No. 2 (Plate III.) to the meridian 0°–180°. Next, if the orbit-pole be at one of the points A, that is, if the orbit-plane pass through a ray though the position of the orbit-pole be here determinate, yet as there is a series of orbits, this also will give rise to a pianogram: I call it Pianogram No. 3. The orbit-pole may pass along a separator circle (viz. the orbit-plane be parallel to a ray), this is Pianogram No. 4. And, lastly, the orbit-pole may pass along the ecliptic (or the orbit-plane may pass through the axis SZ), I call this Pianogram No. 5. But the last three planograms are not considered in the like detail as the first two, and I have not, in regard to them, tabulated the results, nor given any Plates.

[p. 430]

Article Nos. 66 *to* 82. *Planogram No.* 1, *the Meridian* 90°–270° *(see Plate II.).*

66. Supposing that the orbit-plane rotates about the axis $S1$ (fig. 6, see No. 64) in the plane of the ecliptic, the orbit-pole will describe the meridian 90°–270°, the position of the orbit-pole being $b = 90°$, $c = 0°$ to 90°, or else $b = 270°$, $c = 0°$ to 90°. But the same analytical formula extends to the two half meridians, viz., we may take $b = 90°$, and extend c over 180°, in the final results making c an arc between 0° and 90°, and $b = 90°$, or $= 270°$, as the case requires.

67. Assuming then $b = 90°$, we have

$$\begin{aligned} \alpha,\ \beta,\ \gamma &= 1, && 0, && 0, \\ \alpha',\ \beta',\ \gamma' &= 0, && -\cos c, && \sin c, \\ \alpha'',\ \beta'',\ \gamma'' &= 0, && \sin c, && \cos c, \end{aligned}$$

and, moreover, $x' = x$, $y' = y$: so that instead of $(x'_1,\ y'_1)$, &c., we may write at once $(x_1,\ y_1)$, &c. The formulae become

$$\begin{aligned} x_1 \ :\ y_1 \ :\ 1 &= \sqrt{3}\cos c + \sin c \ :\ 0 \ :\ \sin c + \sqrt{3}\cos c, \\ x_2 \ :\ y_2 \ :\ 1 &= -\sqrt{3}\cos c - 2\sin c \ :\ -3 \ :\ \sin c - 2\sqrt{3}\cos c, \\ x_3 \ :\ y_3 \ :\ 1 &= \ \ \sqrt{3}\cos c - 2\sin c \ :\ -3 \ :\ \sin c - 2\sqrt{3}\cos c, \end{aligned}$$

that is

$$x_1 = 1, \qquad y_1 = 0.$$

(viz. the orbit-plane, as is evident, meets the ray 1 in a fixed point, its intersection with the plane of xy)

$$x_2 = \frac{-\sqrt{3}\cos c - 2\sin c}{\sin c - 2\sqrt{3}\cos c}, \qquad y_2 = \frac{-3}{\sin c - 2\sqrt{3}\cos c},$$

$$x_3 = \frac{\sqrt{3}\cos c - 2\sin c}{\sin c - 2\sqrt{3}\cos c} = -x_2, \qquad y_3 = \frac{-3}{\sin c - 2\sqrt{3}\cos c} = -y_2.$$

[Note: in the printed text the formula for x_3 is given with an obvious typographical sign error; the corrected expression makes $x_3 = -x_2$ and $y_3 = y_2$ symmetric, but the printed page gives $y_3 = -y_2$ which we have preserved verbatim.]

and writing

$$\frac{2\sqrt{3}}{\sqrt{13}}\cos\omega = \frac{1}{\sqrt{13}}, \qquad \frac{1}{\sqrt{13}}\sin\omega = \frac{1}{2\sqrt{3}}\tan\omega,$$

(whence $\omega = 16°\ 6'$) we find

$$\begin{aligned} x_2 &= -\tfrac{1}{2} + \tfrac{3\sqrt{3}}{\sqrt{13}}\tan(c+\omega), \\ y_2 &= +\tfrac{3}{\sqrt{13}}\sec(c+\omega), \end{aligned}$$

and we thence have for the hyperbola, the locus of $(x_2,\ y_2)$ and $(x_3,\ y_3)$

$$\left(x + \tfrac{1}{2}\right)^2 = \tfrac{3}{12}\left(y^2 - \tfrac{3}{4}\right),$$

[p. 431]

viz. the points $(x_2,\ y_2)$ and $(x_3,\ y_3)$ are situate on the hyperbola, symmetrically on opposite sides of the axis of x. For $c = 0$, we have $x_2 = -\frac{1}{2}$, $y_2 = \frac{1}{2}\sqrt{3}$, $(x_2^2 + y_2^2 = 1)$, and the hyperbola at this point touches the circle $x^2 + y^2 = 1$; and similarly for x_3, y_3. The inclination of the asymptotes to the axis of y is given by $\tan\eta = \sqrt{\frac{4}{9}}$, $\eta = 22°\ 56'$.

68. The orbits are conics, focus S and vertex 1. It will be convenient to consider c as passing from $0°$ to $90° - \omega$, and from $0°$ to $-(90° + \omega)$; that is, from $0°$ to $73°\ 54' - \epsilon$, and from $0°$ to $-106°\ 6' + \epsilon$, if ϵ be indefinitely small: the point 2 will thus traverse the upper branch (alone shown in the Plate) of the guide-hyperbola, viz., for $c = 0°$ it will be at the point of contact with the circle; for $c = 73°\ 54' - \epsilon$ it will be at ∞, and for $c = -106°\ 6' + \epsilon$ at ∞'. For $c = 0°$ the orbit is the circle; as c increases positively, it becomes an ellipse of increasing eccentricity and major axis, until for a certain value ($c = 46°\ 48'$ as will appear) it becomes a parabola; it then becomes a hyperbola (concave branch); for $c = 52°\ 45'$ it becomes the hyperbola Σ' subsequently referred to; and for $c = 60°$ (the point 2 being then on the line shown in the figure) the orbit becomes this right line. As c continues to increase, the orbit becomes a hyperbola (convex branch); and ultimately for $c = 73°\ 54' - \epsilon$, the point 2 goes to ∞, and the orbit becomes a hyperbola (convex) Σ, having an asymptote parallel to that of the guide-hyperbola: the inclination to the axis of x being thus $90° - 22°\ 56' = 67°\ 4'$.

69. Next as c increases negatively, the point 2 moves from the point of contact in the other direction to ∞': for $c = 0°$ the orbit is of course the circle, and as c increases negatively the orbits are at first the very same series of orbits as those belonging to the positive values[1], viz., they are first ellipses, of increasing eccentricity and major axis; then for $c = -92° \, 54'$ the orbit is the parabola; the orbits are then hyperbolas (concave), and finally for $c = -106° \, 6' + \epsilon$, when 2 is at ∞', the orbit is a hyperbola Σ', the asymptote of which is parallel to that of the guide-hyperbola, viz., the inclination to the axis of x is $67° \, 4'$.

70. It will be observed that the orbits from the circle to the hyperbola Σ' each intersect the guide-hyperbola (that is, the branch shown in the figure) in two points, the one corresponding to a positive, the other to a negative value of c; in the positive series, the remaining orbits from the hyperbola Σ', through the right line to the convex hyperbola Σ, each intersect the guide-hyperbola (same branch) in a single point only, for which c is positive.

71. There is, in the passage of the orbit-pole from $c = -106° \, 6' + \epsilon$ to $c = 73° \, 54' - \epsilon$, say at $c = 73° \, 54'$, a discontinuity of orbit, viz., an abrupt change from the concave hyperbola Σ' to the concave hyperbola Σ; observe that the direction of the asymptotes being the same in each, the eccentricity e has the same value.

[p. 432]

The point in question ($b = 90°$, $c = 73° \, 54'$) is one of the points B of the spherogram, and the hyperbolas Σ, Σ' are two of the four orbits belonging to this point. And, by what precedes, it appears that as the orbit-pole passes through this point along a meridian downwards to the ecliptic the change is from a concave to a convex orbit.

72. On account of the symmetry in regard to the axis of x, the equation of the orbit will be of the form $r = \dfrac{Ax + B}{1}$; viz., the equation is at once found to be

$$r = \frac{x_2 - 1}{x_2 - 1}\,(x - 1).$$

73. The eccentricity is the coefficient A taken positively ($e = \pm A$): it is in the present case proper to attend to the value of the coefficient itself,

$$A = \frac{r_2 - 1}{x_2 - 1},$$

the sign of A will then indicate the position of the centre of the orbit, viz., according as A is positive or negative the centre will be on the negative or the positive side of the focus S. To investigate the variation of A, we may express it as a function of $\tan c$, $= \lambda$ suppose. We have

$$x_2 = \frac{\sqrt{3} - 2\lambda}{\lambda - 2\sqrt{3}}, \qquad y_2 = \frac{-3}{\lambda - 2\sqrt{3}},$$

and thence

$$r_2 = \frac{1}{\lambda - 2\sqrt{3}}\,R_2, \qquad R_2 = \pm\sqrt{12 - 4\sqrt{3}\lambda + 13\lambda^2},$$

viz., r_2 must be positive, that is, R_2 is positive or negative according to the sign of $\lambda - 2\sqrt{3}$; negative if $\lambda < 2\sqrt{3}$ or $c < 73° \, 54'$, positive if $\lambda > 2\sqrt{3}$ or $c > 73° \, 54'$. And we have then

$$A = \frac{\lambda - 2\sqrt{3} - R_2}{3(\lambda - \sqrt{3})}.$$

But a more convenient formula is obtained by writing

$$\theta = -\cot c + \frac{1}{2\sqrt{3}}, \qquad a = \frac{1}{2\sqrt{3}} - 1,$$

[1]Of course, as corresponding to different values of c, they are not the same orbits in space, but they are only the same curves in the planogram.

we then have

$$\sqrt{1+\theta^2} = \theta r_2,$$

which determines the sign of the radical, viz., this must have the same sign as θ; and then for the coefficient

$$A = \frac{2}{3(\theta+a)}\big(-\sqrt{1+\theta^2}+\theta\big).$$

[p. 433]

74. For c a small arc $= \epsilon$, θ is large and negative, and $\sqrt{1+\theta^2}$, having the same sign as θ, is $= \theta$ nearly; we have therefore

$$A = \frac{2}{3\theta}\cdot\frac{-1}{2\theta} = \frac{-1}{3\theta^2} \quad \text{approximately.}$$

For c nearly $= 60°$, say $c = 60° \pm \epsilon$,

$$\cot c = \cot 60° \pm \epsilon \csc^2 60° = \tfrac{1}{\sqrt{3}} \pm \frac{4\epsilon}{3},$$

$$\theta = -\frac{1}{2\sqrt{3}} \pm \frac{4\epsilon}{3}, \qquad \theta + a = \pm\frac{4\epsilon}{3}, \qquad \sqrt{1+\theta^2} = -\frac{\sqrt{13}}{2\sqrt{3}},$$

and thence

$$A = \frac{-1+\sqrt{13}}{2\sqrt{3}} \pm \frac{4\epsilon}{3} = \pm\frac{\sqrt{3}}{8}\cdot\frac{-1+\sqrt{13}}{\epsilon},$$

viz., this is $-\infty$ for $c = 60° - \epsilon$, and $+\infty$ for $c = 60° + \epsilon$.

For c nearly $= 90° - \omega$, say first $c = 73°\,54' - \epsilon$, we have

$$\cot c = \frac{1}{2\sqrt{3}} + \tfrac{13}{12}\epsilon, \qquad \theta = -\tfrac{13}{12}\epsilon, \qquad \theta + a = -\frac{1}{2\sqrt{3}}, \qquad \sqrt{1+\theta^2} = -1,$$

whence

$$A = \frac{4}{\sqrt{3}} = 2{\cdot}30940;$$

but if $c = 73°\,54' + \epsilon$, then $\theta = \frac{13}{12}\epsilon$, $\theta + a = -\dfrac{1}{2\sqrt{3}}$, $\sqrt{1+\theta^2} = +1$, and

$$A = -\frac{4}{\sqrt{3}} = -2{\cdot}30940,$$

viz., there is an abrupt change from $A = +\dfrac{4}{\sqrt{3}}$ to $A = -\dfrac{4}{\sqrt{3}}$; corresponding to the discontinuity of orbit already referred to. We may diminish c by $180°$, and consider the last-mentioned value, $A = -\dfrac{4}{\sqrt{3}}$, as belonging to $c = 90° - \omega + \epsilon = -(106°\,6' - \epsilon)$.

75. Consider next that c passes from $0°$ to $-(106°\,6' - \epsilon)$. First if c is a small negative quantity $c = -\epsilon$, θ is large and positive, and $\sqrt{1+\theta^2}$ having the same sign as θ (positive) is $= \theta + \dfrac{1}{2\theta}$ nearly, we have therefore $A = \dfrac{2}{3\theta}\cdot\dfrac{-1}{2\theta} = \dfrac{-1}{3\theta^2}$ (same as for $c = +\epsilon$). And it is easy to see that as c increases negatively, A is always increasing negatively, its value for $c = -90°$ being $A = \dfrac{1-\sqrt{13}}{3} = -{\cdot}8685$, and for $c = -106°\,6' + \epsilon$ A being $= -2{\cdot}30940$ as above. We have a diagram of A (see next page).

[p. 434]

76. It thus appears that from $A = -\dfrac{4}{\sqrt{3}}$ to $A = -\dfrac{4}{\sqrt{3}}$, there are always to any given value of A two values of c, or positions of the orbit-pole. In particular if A be $= -1$, the curve will be a parabola; the values of c lying between $0°$, $60°$ and between $73° \, 54'$, $90°$ respectively.

[Figure 7: schematic graph of A as a function of c, showing the asymptotes at $c = 60°$ and at $c = -106° \, 6'$, the level $A = -1$ marked along the horizontal axis, with the abscissa scale running from $-100°$ on the left through $0°$ at centre to $90°$ on the right.]

To find them, writing $A = -1$, we have

$$-3\theta - 3a = 2\theta - 2\sqrt{1+\theta^2}, \quad \text{that is,} \quad 5\theta + 3a = 2\sqrt{1+\theta^2},$$

or

$$21\theta^2 + 30a\theta + 9a^2 - 4 = 0,$$

that is, substituting for a its value $= \dfrac{1}{2\sqrt{3}}$,

$$21\theta^2 + 5\sqrt{3}\theta - \tfrac{13}{4} = 0, \qquad (14\theta\sqrt{3} + 5)^2 = 116,$$

or

$$\theta = \frac{-5 \pm \sqrt{116}}{14\sqrt{3}},$$

giving

$$\theta = -{\cdot}23797, \quad \text{or} \quad \theta = -{\cdot}65034,$$
$$\cot c = +{\cdot}93902, \quad \text{or} \quad \cot c = +{\cdot}05071,$$
$$c = 46° \, 48', \quad \text{or} \quad c = 87° \, 6'.$$

77. It has been seen that $c = 73° \, 54' + \epsilon$ gives $A = -\dfrac{4}{\sqrt{3}} = -2{\cdot}30940$; there will be between $0°$ and $60°$ another value of c, giving for A this same value; to find this value write $A = -\dfrac{4}{\sqrt{3}}$, then we have

$$2\theta\Big(1 + \frac{2}{\sqrt{3}}\theta\Big) - 4 = \frac{4}{\sqrt{3}}\Big(\sqrt{1+\theta^2} - \theta\Big) \cdot \sqrt{3},$$

[p. 435]

that is

$$(1 + 2\sqrt{3})\theta + 1 = \sqrt{1+\theta^2},$$

or

$$\theta^2(12 + 4\sqrt{3}) + (2 + 4\sqrt{3})\theta = 0,$$

satisfied as it should be by $\theta = 0$, and also by

$$\theta = -\frac{1}{2(3+\sqrt{3})},$$

giving

$$\cot c = {\cdot}76038 \quad \text{or} \quad c = 52° \, 45'.$$

78. Representing the equation of the orbit by

$$r = Ax \pm a(1 - A^2),$$

we have for the point 1,

$$1 = A \pm a(1 - A^2),$$

that is

$$a = \frac{\pm 1}{1 + A},$$

where the sign is to be taken so that a shall be positive.

79. With a view to the calculation of the times of passage, I calculate a series of values of x_2, y_2, r_2, A, a, for values of c at the intervals of 5° and for a few intermediate values; we have x_3, y_3, $r_3 = x_2$, y_2, r_2, so that these are known; so long as the orbit is an ellipse, the time of passage between the points 2 and 3, say T_{23}, may be calculated by Lambert's equation, the length of the chord $= y_2 - y_3, = 2y_2$ being known without any fresh calculation. And then the times T_{12} and T_{31} being equal, and the sum $T_{12} + T_{23} + T_{31}$ being equal to the whole periodic time (reckoned as $= 3a^{\frac{3}{2}}$), the times T_{12} and T_{31} are also known. But when the orbit is a concave hyperbola there is no time T_{23}, and the other two times $T_{12}, = T_{31}$ must be calculated. For the reason referred to (ante, No. 39) I did not use Lambert's equation, and it was less necessary to do so, by reason that, the transverse axis coinciding with the axis of x, the other formula could be employed without difficulty.

80. The formulae for x_2, y_2 adapted to logarithmic calculation are

$$\log(x_2 + \cdot 615.39) = 11\cdot 60174 + \log\tan(c + 16^\circ\, 6'),$$
$$\log y_2 = n\cdot 92015 + \log\sec(c + 16^\circ\, 6'),$$

where y_2 is always positive, but the sign of x_2 must be attended to. The values of r_2 and its inclination ϕ_2 to the axis of x are then to be calculated from

$$\frac{y_2}{x_2} = \tan\phi_2; \quad r_2 = x_2 \sec\phi_2 \text{ or } = y_2 \csc\phi_2,$$

(viz. for r_2 it is proper to use the first or the second value, according as x_2 is greater or less than y_2).

[p. 436]

We have then $e\,(= \pm A)$ and a from the foregoing formulae

$$A = \frac{r_2 - 1}{x_2 - 1}, \qquad a = \frac{\pm 1}{1 + A},$$

where a, e are each of them positive.

And then for the Times

$$T_{31} = T_{12} = \tfrac{3}{2\pi}\, a^{\frac{3}{2}} \left(\chi' - \chi - \sin\chi' + \sin\chi\right); \qquad (\log \tfrac{3}{2\pi} = \bar{1}\cdot 67894)$$

where

$$a\cos\chi = a - r_2 - y_2,$$
$$a\cos\chi' = a - r_2 + y_2,$$

and attention is necessary in order to the selection of the proper values of the angles χ, χ'.

And finally

$$T_{23} = T_{12} - \tfrac{3}{2\pi}\,(3a^{\frac{3}{2}} - T_{12}).$$

81. I subjoin a specimen; the characteristics of the logarithms are (as in the actual calculations) omitted.

[Specimen calculation, $b = 90°$, $c = 20°$. The page presents a column-arithmetic worked example. The numerical content from the scan is reproduced below in alignment.]

$b = 90^\circ \quad c = 20^\circ$
$c + 16^\circ\,6' = 36^\circ\,6'$

$\log\sec$ ·09259	$\log\tan$ ·86285
·92015	·60174
·01274	·46459
$y_2 = 1{\cdot}0297$	·61539
	·01274
	·29147
	·51044
	$\therefore x_2 = -{\cdot}32{\cdot}392$
	$\log = {\cdot}51044$
	·02046
	·01274
·50230	·03320
$\phi_2 = 72^\circ\,33'$	$r_2 = 1{\cdot}0794$
·0794 $\log = \bar{1}{\cdot}89982$	·94003 $\log = \bar{1}{\cdot}97314$
1·3239 $\log = {\cdot}12185$	comp $= {\cdot}02686$
	·77797

[p. 437]

$A = -{\cdot}059975 \qquad a = 1{\cdot}0638$
1·0638
1·0794

$A - r_2 = -{\cdot}0156$
$y_2 = +1{\cdot}0297$
$1{\cdot}0141 = a\cos\chi'$
$-1{\cdot}0453 = a\cos\chi$

·00608	·01924
·02686	·02686
·97922	·99238
$\chi' = 17^\circ\,35'$	$\chi\,(=\text{Supp. } 10^\circ\,42') = 169^\circ\,18' \quad \chi - \chi' = 151^\circ\,43'$

151° 2·63544 ·02686
43′ ·01250 ·01343
$-\sin\chi$ $-$ ·18566 ·47712
$\sin\chi'$ ·30209 ·51741
2·95003 $3a^{\frac{3}{2}} = 3{\cdot}2916$
·18566 1·4482
2·76437 $\log = {\cdot}44160$ 1·8434
·02686 $T_{12} = T_{31} = {\cdot}9217$
·01343
·67894
$T_{23} = 1{\cdot}4482$ ·16083

82. For the Time in a hyperbola, we have

$$T_{31} = T_{12} = \frac{3}{2\pi}\,a^{\frac{3}{2}}\left\{e\tan u - h{\cdot}l.\tan(45^\circ + \tfrac{1}{2}u)\right\},$$

where

$$\tan u = \frac{y_2}{a\sqrt{e^2-1}}.$$

[p. 438]

Taking as a specimen the case $c = 75°$, we have here

$$\begin{array}{lll} a = \cdot 9004 & e = 2\cdot 1106 & y_2 = 43\cdot 341 \\ \log = \bar{1}\cdot 95444 & \log = \cdot 32441 & \log = 1\cdot 63690 \end{array}$$

$$\begin{aligned} a(e^2 - 1) &= 3\cdot 1106 \\ \log &= \cdot 49284 \end{aligned}$$

and then the calculation is

$$\begin{array}{ll}
\log a = \bar{1}\cdot 95444 & \\
\log a(e^2-1) = \cdot 49284 & \\
\quad\overline{}\ \cdot 44728 & \\
\log a\sqrt{e^2-1} = \cdot 22364 & \\
\log y_2 = 1\cdot 63690 & \\
\log\tan u = 1\cdot 41326 & u = 87°\,47' \\
 & h\cdot l\tan(45° + \tfrac{1}{2}u) = 3\cdot 95140 \\
\quad\cdot 32441 & e\tan u = 54\cdot 660 \\
\quad\overline{}\ \cdot 73767 & \quad 3\cdot 951 \\
 & \quad 50\cdot 709 \\
 & \log = \cdot 70508 \\
 & \quad \cdot 95444 \\
 & \quad \cdot 47722 \\
 & \quad \cdot 67894 \\
 & \\
 & \quad 3\cdot 1568 \\
 & T_{12} = T_{31} = 20\cdot 686
\end{array}$$

83. In the case of the parabola $p = 1$, and the expression for the Times is

$$T_{12} = T_{31} = \frac{1}{6\sqrt{\pi}}\Big\{(\rho + \rho' + \gamma)^{\frac{3}{2}} - (\rho + \gamma)^{\frac{3}{2}}\Big\},$$

where for

$$\begin{aligned} c &= 46°\,48' \\ c &= 87°\,6' \end{aligned}$$

we have

$$\begin{aligned} T_{12} = T_{31} &= \cdot 787, \\ T_{12} = T_{31} &= 2\cdot 588. \end{aligned}$$

[p. 439]

Planogram No. 1, first part, $b = 90°$.

	c	x_2	y_2	ϕ_2	r_2	A	a	T_{12}	T_{23}	T_{31}
Circle	0°	−·500	+·866	60°	1·000	0	1·000	1·000	1·000	1·000
	5	·461	·892	62° 39′	1·004	−·003	1·003			
	10	·420	·927	65 38	1·017	−·012	1·012	·960	1·135	·960
	15	·374	·972	68 56	1·041	−·030	1·031			
Ellipses	20	·324	1·030	72 33	1·079	−·060	1·064	·922	1·448	·922
	25	·267	1·104	76 26	1·136	−·107	1·120	·887	2·275	·887
	30	·200	1·200	80 32	1·216	−·180	1·220			
	35	·120	1·325	84 49	1·330	−·295	1·418			
	40	−·021	1·492	90 48	1·492	−·482	1·931	·838	6·371	·838
	40° 54′	·000	1·515	90	1·515	−·515	2·061			
	45	+·109	1·722	93 37	1·725	−·814	5·362			
Parab.	46° 48′	·166	1·826	95 11	1·834	−1·000	∞	·787	∞	·787
	50	·287	2·054	97 57	2·074	−1·505	1·981	·750	$\sim$	·750
Hyperbs. {	52° 45′	·418	2·306	100 16	2·344	−2·309	·764	·628	$\sim$	·628
	55	·552	2·569	102 8	2·627	−3·632	·380			
Line	60	1·000	3·464	106 6	3·605	$\mp\infty$	·000	·000	$\sim$	·000
	65	1·937	5·378	109 48	5·716	+5·032	·166		Convex orbit.	
Convex	70	5·248	12·233	113 13	13·311	+2·898	·257			
	$73° 54' - \epsilon$	$+\infty$	∞	115 39	∞	+2·309	·302			
	$73° 54' + \epsilon$	$-\infty$	∞	64 21	∞	−2·309	·764	∞	$\sim$	∞
	76	−21·432	43·341	63 42	48·346	−2·111	·900	20·68	$\sim$	20·68
Hyperbs.	80	−4·356	7·830	60 55	8·960	−1·486	2·056	3·856	$\sim$	3·856
	85	−2·653	4·322	58 27	5·072	−1·115	8·718	2·912	$\sim$	2·912
Parab.	87° 6′	−2·320	3·644	57 31	4·320	−1·000	∞	2·588	∞	2·588
Ellipse	90	−2·000	3·000	56 18	3·606	−·869	7·622	2·255	58·62	2·255

The mark $\sim$ in the T_{23} column shows that there is no Time T_{23}.

[p. 440]

Planogram No. 1, second part, $b = 270°$.

	c	x_2	y_2	ϕ_2	r_2	A	a	T_{12}	T_{23}	T_{31}
Circ.	0°	−·500	+·866	60°	1·000	0	1·000	1·000	1·000	1·000
	5	·537	·848	57 39	1·003	−·002	1·002			
	10	·573	·837	55 37	1·014	−·009	1·009	1·044	·951	1·044
	15	·608	·832	53 52	1·030	−·019	1·019			
	20	·643	·834	52 23	1·053	−·032	1·033	1·091	·969	1·091
	25	·678	·842	51 10	1·081	−·048	1·051			
	30	·714	·857	50 11	1·116	−·068	1·073	1·145	1·043	1·145
All Ellipses	35	·752	·879	49 27	1·157	−·090	1·098			
	40	·793	·910	48 51	1·207	−·115	1·130	1·207	1·192	1·207
	45	·836	·950	48 40	1·266	−·145	1·169			
	50	·884	1·002	48 36	1·336	−·179	1·217	1·283	1·464	1·283
	55	·938	1·069	48 45	1·442	−·218	1·278			
	60	1·000	1·154	49 7	1·527	−·264	1·358	1·377	1·983	1·377
	65	1·074	1·266	49 42	1·660	−·318	1·466			
	70	1·164	1·412	50 31	1·830	−·383	1·622	1·506	3·036	1·506
	75	1·280	1·611	51 35	2·056	−·464	1·864			
	80	1·431	1·891	52 53	2·372	−·564	2·295	1·771	6·888	1·771
	85	1·651	2·311	54 28	2·840	−·694	3·269			
	90	−2·000	+3·000	56 18	3·606	−·869	7·622	2·255	58·62	2·255

[p. 441]

Article Nos. 84 to 94. Planogram No. 2, the Meridian 0°–180° (see Plate III.).

84. The orbit-plane here rotates about an axis in the plane of the ecliptic at right angles to $S1$ (Fig. 6). The entire meridian is given by $b = 0°$, $c = 0°$ to $90°$, and $b = 180°$, $c = 0°$ to $90°$, but it is sufficient to consider one of these half meridians, say the latter of them, as the series of values is the same for each of them, with only an interchange of the points 2, 3. I write therefore, $b = 180°$, so that we have

$$\begin{aligned} \alpha,\ \beta,\ \gamma &= 0,\ 1,\ 0, \\ \alpha',\ \beta',\ \gamma' &= -\cos c,\ 0,\ -\sin c, \\ \alpha'',\ \beta'',\ \gamma'' &= -\sin c,\ 0,\ \cos c, \end{aligned}$$

consequently

$$\begin{aligned} x_1' : y_1' : 1 &= \sin c \ :\ -\sqrt{3}\ :\ \sqrt{3}\cos c, \\ x_2' : y_2' : 1 &= -1 - 3\cos c - 2\sin c\ :\ -\sqrt{3}\ :\ -\sqrt{3}\sin c - 2\sqrt{3}\cos c, \\ x_3' : y_3' : 1 &= -1 - 3\cos c - 2\sin c\ :\ -\sqrt{3}\ :\ \sqrt{3}\sin c - 2\sqrt{3}\cos c, \end{aligned}$$

and moreover $x = y'$, $y = -x'$; so that, introducing into the formulae (x_1, y_1), &c., in place of the (x_1', y_1'), &c., we have

$$x_1 = \sec c, \qquad y_1 = -\tfrac{1}{\sqrt{3}}\tan c,$$

$$x_2 = \frac{-1}{\sin c - 2\cos c}(\sin c + 2\cos c) + \tfrac{1}{\sqrt{3}} \cdot \frac{1 + 2\sin c - 3\cos c}{\sqrt{3}\sin c - 2\cos c},$$

$$x_3 = \frac{-1}{\sin c - 2\cos c}(\sin c + 2\cos c) - \tfrac{1}{\sqrt{3}} \cdot \frac{1 - 2\sin c - 3\cos c}{\sqrt{3}\sin c - 2\cos c},$$

which, putting

$$\cos\delta = \tfrac{2}{\sqrt{5}}, \quad \sin\delta = \tfrac{1}{\sqrt{5}}, \quad \tan\delta = \tfrac{1}{2}, \quad \delta = 26°\,34',$$

become

$$\begin{aligned} x_1 &= \sec c, & y_1 &= -\tfrac{1}{\sqrt{3}}\tan c, \\ x_2 &= -\tfrac{1}{\sqrt{5}}\sec(c - \delta), & y_2 &= \tfrac{1}{\sqrt{3}}\left\{\tfrac{2}{5} + \tfrac{1}{2}\tan(c - \delta)\right\}, \\ x_3 &= -\tfrac{1}{\sqrt{5}}\sec(c + \delta), & y_3 &= \tfrac{1}{\sqrt{3}}\left\{-\tfrac{2}{5} + \tfrac{1}{2}\tan(c + \delta)\right\} \end{aligned}$$

so that the guide-hyperbolas are (angle of asymptotes $= 30°$):

$$\begin{aligned} &x_1^2 - 3y_1^2 = 1, \\ &\quad x_2^2 = 15y_2^2 - 16\sqrt{3}\,y_2 + 13, \qquad \tan^{-1}\tfrac{1}{\sqrt{15}} = 14°\,28' \\ &\quad x_3^2 = 15y_3^2 + 16\sqrt{3}\,y_3 + 13, \end{aligned}$$

[p. 442]

It is easy to verify that
Hyperbola 2 passes through $x_2 = -\frac{1}{2}$, $y_2 = +\frac{1}{2}\sqrt{3}$, and touches there circle $x^2 + y^2 = 1$,
3 $x_3 = -\frac{1}{2}$, $y_3 = -\frac{1}{2}\sqrt{3}$ „
and we thus have the figure in the Plate.

85. The figure shows the motion of the points 1, 2, 3, along their respective hyperbolas, viz. $c = 0°$ to $90°$, the point 1 moves from contact with the circle, along a half branch to infinity: 2 moves from contact along a small portion of the half branch; 3 moves from contact, along the half branch to infinity for $c = \tan^{-1} 2 = 63°\,26'$, and then reappearing at the opposite infinity, as c increases to $90°$ describes a portion of the opposite half branch.

86. For $c = 0$, the orbit is the circle; as c increases the orbit becomes elliptic, then parabolic, $c = 51°$, and afterwards hyperbolic (concave); until for $c = 60°$, the three points are on the horizontal line of the figure, and the orbit is this right line; it is to be noticed that the arrangement of the points on these orbits is 1, 2, 3; so that for the parabola, T_{23} is $= \infty$, and for the hyperbolas and right line T_{23} does not exist.

87. For $c < 60^\circ$ until $c = 63^\circ\,26'$ the orbit is a convex hyperbola, the arrangement of the points being still 1, 2, 3: say for $c = 63^\circ\,26' - \epsilon$, the orbit is the convex hyperbola Ω. At $c = 63^\circ\,26'$ there is an abrupt change of orbit; say for $c = 63^\circ\,26' + \epsilon$ the orbit is a concave hyperbola Ω_1; and for $c = 65^\circ\,52'$ the orbit is a parabola; the arrangement of the points on these orbits is 2, 1, 3; so that for the hyperbolas T_{23} does not exist, and T_{21} is $= \infty$ for the parabola. Observe also that for the hyperbola Ω_1, the point 3 is at infinity, or we have $T_{31} = \infty$. As c continues to increase, the orbit becomes an ellipse, the eccentricity having a minimum value $= \cdot 628$ (about), for $c = 69^\circ$ (about). For $c = 89^\circ\,20'$ the orbit is again a parabola, and then until $c = 90^\circ$ it is a hyperbola; the order of the points on the last-mentioned parabola and hyperbolas being 1, 3, 2; so that for the parabola T_{32} is $= \infty$, and for the hyperbolas T_{32} does not exist. In the hyperbola for $c = 90^\circ$, say the hyperbola Ω', the point 1 is at infinity, or we have $T_{12} = \infty$. The foregoing results, obtained (except as to the numerical values) by consideration of the figure, will be confirmed by means of the calculated values of e.

88. The equation of the orbit may be written

$$\begin{vmatrix} \dfrac{r}{\sqrt{3}} & \dfrac{x}{\sqrt{3}} & y & \dfrac{1}{\sqrt{3}} \\ r_1 \cos c & ,\quad 1, & \sin c & \cos c \\ +r_2(\sin c + 2\cos c), & -1, & +2\sin c - 3\cos c, & \sin c + 2\cos c \\ -r_3(\sin c + 2\cos c), & 1, & +2\sin c - 3\cos c, & \sin c - 2\cos c \end{vmatrix} = 0,$$

[p. 443]

or developing, this is

$$\begin{aligned}
&\frac{r}{\sqrt{3}}\,6(\sin^2 c - 3\cos^2 c)\\
&-\frac{x}{\sqrt{3}}\Big\{\,4r_1(\sin^2 c - 3\cos^2 c)\cos c\\
&\qquad\qquad - r_2(\sin c + 2\cos c)(\sin^2 c - 3\cos^2 c)\\
&\qquad\qquad + r_3(\sin c - 2\cos c)(\sin^2 c - 3\cos^2 c)\Big\}\\
&+y\,\{\,-2r_1\sin c\cos c\\
&\qquad\qquad + r_2(-\sin^2 c + \sin c\cos c + 6\cos^2 c)\\
&\qquad\qquad + r_3(\,\sin^2 c + \sin c\cos c - 6\cos^2 c)\}\\
&-\frac{1}{\sqrt{3}}\Big\{\,r_1\cdot 6\cos^2 c\\
&\qquad\qquad + 3r_2(\sin^2 c + \sin c\cos c + 2\cos^2 c)\\
&\qquad\qquad + 3r_3(\sin^2 c - \sin c\cos c - 2\cos^2 c)\Big\} = 0\,;
\end{aligned}$$

(observe that the orbit will be a right line if $\sin^2 c - 3\cos^2 c = 0$, that is if $c = 60°$, which is right, since $60°$ is the angular radius of the regulator circle).

89. Putting in the equation $\tan c = \lambda$, and therefore $\cos c = \dfrac{1}{\sqrt{1+\lambda^2}}$, the equation becomes

$$r = \frac{1}{6\sqrt{1+\lambda^2}}\big(4r_1 - (\lambda+2)r_2 + (\lambda-2)r_3\big)x + \frac{1}{2\sqrt{3}(\lambda^2-3)}\big(2\lambda r_1 + (\lambda-3)(\lambda-2)r_2 - (\lambda+3)(\lambda+2)r_3\big)y + \frac{1}{2(\lambda^2-3)}\big(-2r_1 + (\lambda-1)(\lambda+2)r_2 + (\lambda+1)(\lambda-2)r_3\big).$$

We have

$$x_1 = \sqrt{1+\lambda^2},\quad x_2 = \frac{-\sqrt{1+\lambda^2}}{\lambda+2},\quad x_3 = \frac{-\sqrt{1+\lambda^2}}{\lambda-2},$$
$$y_1 = \tfrac{1}{\sqrt{3}}\,\lambda,\quad y_2 = \tfrac{1}{\sqrt{3}}\cdot\frac{2\lambda+3}{\lambda+2},\quad y_3 = \tfrac{1}{\sqrt{3}}\cdot\frac{2\lambda-3}{\lambda-2},$$

and thence, writing for shortness

$$\begin{aligned}
R_1 &= \sqrt{1+\tfrac{4}{3}\lambda^2},\\
R_2 &= \tfrac{1}{\sqrt{3}}\sqrt{7\lambda^2+12\lambda+12},\\
R_3 &= \tfrac{1}{\sqrt{3}}\sqrt{7\lambda^2-12\lambda+12},
\end{aligned}$$

[p. 444]

we have

$$r_1 = R_1, \quad r_2(\lambda+2) = R_2, \quad r_3(\lambda-2) = R_3,$$

where r_1, r_2, r_3 are positive, and the signs of R_1, R_2, R_3 must be determined accordingly; viz., R_1 is always positive, and ($c = 0°$ to $c = 90°$, as here supposed) R_2 is also positive but R_3 has the same sign as $\lambda - 2$; viz., $c = 0°$ to $c = 63° \, 26'$, R_3 is negative; and $c = 63° \, 26'$ to $c = 90°$, R_3 is positive. It is to be observed that this position, $c = \tan^{-1} 2 = 63° \, 26'$, of the pole is the intersection of the meridian $b = 180°$ by a separator circle, and corresponds to an intersection at infinity on the ray 3.

90. Substituting the foregoing values of r_1, r_2, r_3, the equation of the orbit becomes

$$r = \frac{1}{6\sqrt{1+\lambda^2}}(4R_1 - R_2 + R_3)x + \frac{1}{2\sqrt{3}(\lambda^2-3)}\{\lambda(2R_1 + R_2 - R_3) - 3(R_2+R_3)\}\, y + \frac{1}{2(\lambda^2-3)}[\lambda(R_2+R_3) - 2R_1 - R_2 + R_3],$$

where $\lambda = \tan c$; and the equation of the orbit may thence be calculated for any given value of c.

91. The analytical expression for the eccentricity is

$$e = \sqrt{A^2 + B^2},$$

[Figure 8: graph of the eccentricity e versus c, showing the values $e = 0$ at $c = 0°$, the parabolic levels $e = 1$ marked, the vertical asymptote near $c = 63° \, 26'$ between the convex and concave hyperbola branches, the minimum near $c = 69°$ at $e \approx {\cdot}628$, and the second parabolic level near $c = 89° \, 20'$. Abscissa runs $0°$ to $90°$ with 1-unit ordinate marks.]

where, as above,

$$A = \frac{1}{6\sqrt{1+\lambda^2}}\big(4R_1 - R_2 + R_3\big),$$
$$B = \frac{1}{2\sqrt{3}(\lambda^2-3)}\big\{\lambda(2R_1 + R_2 - R_3) - 3(R_2+R_3)\big\};$$

[p. 445]

but this expression is too complicated to allow of an analytical discussion of the series of values of e (such as was given for A, ϵ, in planogram No. 1). The numerical calculation gives the results mentioned ante No. 87, viz., $c = 0$, $e = 0$; $c = 51°$, $e = 1$; $c = 60°$, $e = \infty$; $c = 63° \, 26' - \epsilon$, $e = 4{\cdot}912$; $c = 63° \, 26' + \epsilon$, $e = 1{\cdot}853$; $c = 69°$, $e = {\cdot}628$ (min.); $c = 89° \, 20'$ (viz. $\lambda = 86{\cdot}176$), $e = 1$; $c = 90°$, $e = 1{\cdot}018$; values which are exhibited in the diagram in the preceding page.

92. It may be further remarked, in reference to the formula $r = Ax + By + C$, that for $c = 60°$, that is $\lambda = \sqrt{3}$, we have A and C each infinite, but B finite, and of opposite signs; viz., the equation becomes $r = -{\cdot}2242\, x + \infty(y-1)$, that is $y = 1$, orbit a right line as above.

The abrupt change at $c = 63° \, 26'$, $\lambda = 2$, arises from the change of sign of R_3; viz., $c = 63° \, 26' - \epsilon$, $R_3 = -2{\cdot}309$, but $c = 63° \, 26' + \epsilon$, $R_3 = +2{\cdot}309$; the two orbits are

$$\begin{array}{llll} c = 63° \, 26' - \epsilon, & r = +{\cdot}234\, x + 4{\cdot}906\, y - 3{\cdot}671, & e = 4{\cdot}912, & a = {\cdot}159, \\ c = 63° \, 26' + \epsilon, & r = {\cdot}578\, x - 1{\cdot}761\, y + 3{\cdot}257, & e = 1{\cdot}853, & a = 1{\cdot}338. \end{array}$$

For $c = 90°$ the equation is

$$r = {\cdot}770\, x + {\cdot}666\, y + 1{\cdot}527$$

and therefore $e = \sqrt{\ } = 1{\cdot}018$ as above; $a = \frac{9}{2}\sqrt{21} = 41{\cdot}243$.

It is to be added that for c nearly $= 90°$, or λ very large, we have

$$R_1 = \tfrac{2}{\sqrt{3}}\lambda, \quad R_2 = \sqrt{\tfrac{7}{3}}\lambda + 2\sqrt{3}, \quad R_3 = \sqrt{\tfrac{7}{3}}\lambda - 2\sqrt{3},$$

and thence

$$A = \tfrac{4}{3\sqrt{3}} - \tfrac{2}{\sqrt{21}} \cdot \frac{1}{\lambda} = {\cdot}770 - {\cdot}430\frac{1}{\lambda},$$
$$B = \tfrac{2}{\sqrt{7}}\frac{1}{\lambda} - \tfrac{6}{\sqrt{3}}\frac{1}{\lambda^2} = {\cdot}666 - 1{\cdot}890\frac{1}{\lambda^2},$$
$$C = \sqrt{\tfrac{7}{3}} - \tfrac{4}{\sqrt{3}}\frac{1}{\lambda} = 1{\cdot}527 - 1{\cdot}555\frac{1}{\lambda}.$$

It was, in fact, by means of these expressions that the value $\lambda = 86{\cdot}176$ ($c = 89^\circ\,20'$) corresponding to the last-mentioned parabolic orbit was obtained.

[p. 446]

93. For the calculation of the table we have

$$\begin{aligned}\log x_1 &= \overline{10} + \log\sec c,\\ \log y_1 &= \overline{10}{\cdot}76144 + \log\tan c,\\ \log x_2 &= \overline{10}{\cdot}65052 + \log\sec(c - 26^\circ\,34'),\\ \log(y_2 - {\cdot}92376) &= \overline{10}{\cdot}06247 + \log\tan(c - 26^\circ\,34'),\\ \log x_3 &= \overline{10}{\cdot}65052 + \log\sec(c + 26^\circ\,34'),\\ \log(y_3 + {\cdot}92376) &= \overline{10}{\cdot}06247 + \log\sec(c + 26^\circ\,34'),\end{aligned}$$

the values of r_1, r_2, r_3, are then calculated from

$$x_1 = r_1\cos\phi_1, \quad y_1 = r_1\sin\phi_1,$$

or say

$$\frac{y_1}{x_1} = \tan\phi_1, \quad r_1 = x_1\sec\phi_1, \ \&c.$$

and those of the chords γ_{12}, γ_{23}, γ_{31}, from

$$x_1 - x_2 = \gamma_{12}\cos\theta_{12}, \quad y_1 - y_2 = \gamma_{12}\sin\theta_{12},$$

or say

$$\frac{y_1 - y_2}{x_1 - x_2} = \tan\theta_{12}, \quad \gamma_{12} = (x_1 - x_2)\sec\theta_{12}.$$

We have then to find the equation of the orbit $r = Ax + By + C$; this might be done by substituting in the determinant expression the numerical values of x_1, y_1, r_1, x_2, y_2, r_2, and x_3, y_3, r_3, so calculating the result, but I have preferred to employ the formula of No. 90, using only the calculated values of r_1, r_2, r_3; viz. we have

$$r_1 = R_1, \quad r_2(\lambda + 2) = R_2, \quad r_3(\lambda - 2) = R_3.$$

which gives the values of R_1, R_2, R_3. And then we have e, ϖ, a, from the equations

$$A = e\cos\varpi, \quad B = e\sin\varpi, \quad a = \frac{\pm C}{1 - e^2},$$

e and a being each regarded as positive. The times in the elliptic, and parabolic orbits are then calculated from Lambert's equation, as explained in regard to Planogram No. 1, but for the hyperbolic orbits, the other formulae were made use of.

[p. 447]

94. I annex a specimen; the characteristics of the logarithms are omitted.

[$c = 20^\circ$. The columnar specimen calculation gives, in summary, the following intermediate and final results.]

$x_1 = 1{\cdot}06418$	$x_2 = -{\cdot}45017$	$x_3 = -{\cdot}65049$
$y_1 = {\cdot}21014$	$y_2 = +{\cdot}91038$	$y_3 = {\cdot}80171$
$\phi_1 = 11^\circ\,10'$	$\phi_2\,(= 63^\circ\,41') = 116^\circ\,19'$	$\phi_3\,(= 50^\circ\,57') = 230^\circ\,57'$
$r_1 = 1{\cdot}0847$	$r_2 = 1{\cdot}0157$	$r_3 = 1{\cdot}0323.$

The calculation of the equation of the orbit is then as follows:

$$\lambda = \cdot 36397,$$
$$\lambda^2 = \cdot 13248,$$
$$\lambda^2 - 3 = -2\cdot 86752,$$
$$\sqrt{1+\lambda^2} = 1\cdot 06477,$$
$$6\sqrt{1+\lambda^2} = 6\cdot 38862,$$
$$2\sqrt{3}(\lambda^2 - 3) = -9\cdot 93330,$$
$$2(\lambda^2 - 3) = -5\cdot 73504,$$
$$R_1 = 1\cdot 0847, \quad R_2 = +2\cdot 4010, \quad R_3 = -1\cdot 6889,$$
$$4R_1 = 4\cdot 3388, \quad 2R_1 = 2\cdot 1694,$$
$$A = \cdot 038982, \quad B = -\cdot 014265, \quad C = +\cdot 10464,$$
$$\varpi\,(= 20^\circ\,6') = 160^\circ\,52', \quad e = \cdot 04151, \quad 1 - e^2 = \cdot 998277, \quad a = 1\cdot 0481.$$

[p. 448]

[The page is occupied entirely by columnar arithmetic continuing the specimen calculation begun on p. 447. The principal intermediate values are reproduced above; the column totals yield the same $A = \cdot 038982$, $B = -\cdot 014265$, $C = \cdot 10464$, and lead through $e^2 = \cdot 001723$, $1 - e^2 = \cdot 998277$, to $a = 1\cdot 0481$.]

[p. 449]

[The greater part of this page is blank in the original.]

The calculation of the Times is similar to that for the first planogram, and requires no further illustration.

The Table for Planogram No. 2 is as follows:

[pp. 450–451]

Planogram No. 2, $b = 180^\circ$, $c = 0^\circ$ to 90°.

[The original lays the table across two facing pages: a left half (p. 450) carrying columns c, x_1, y_1, x_2, y_2, x_3, y_3, r_1, ϕ_1, r_2, ϕ_2, r_3, ϕ_3; and a right half (p. 451) carrying the Equation of Orbit ($r = Ax + By + C$), the eccentricity e, the angle ϖ, a, the chords γ_{12}, γ_{23}, γ_{31}, and the times T_{12}, T_{23}, T_{31}, repeating c at the right margin. Heading note: "The mark $\sim$ in any of the T columns shows that the Time does not exist." The numerical entries are reproduced below as a single combined table; columns marked all + or all − retain the printed signs.]

	c	x_1	y_1	x_2	y_2	x_3	y_3	r_1	ϕ_1	r_2	ϕ_2	r_3	ϕ_3
Circle	0°	1·000	·000	−·500	+·866	−·500	−·866	1·000	0° 0′	1·000	120° 0′	1·000	240° 0′
	5	1·004	·051	·481	·878	−·525	·853	1·005	2 52	1·001	118 42	1·001	238 40
	10	1·015	·102	·467	·889	·557	·838	1·020	5 44	1·004	117 41	1·006	236 40
	15	1·035	·156	·456	·900	·598	·821	1·047	8 30	1·009	116 54	1·016	233 30
	20	1·064	·210	·450	·910	·650	·802	1·085	11 10	1·016	116 19	1·032	230 30
	25	1·103	·269	·447	·921	·719	·778	1·136	13 43	1·024	115 55	1·060	227 30
Ellipses	30	1·155	·333	·448	·931	·812	·749	1·202	16 6	1·033	115 42	1·104	222 30
	35	1·221	·404	·452	·941	·939	·710	1·286	18 19	1·044	115 40	1·178	217 30
	40	1·305	·484	·460	·951	1·125	·657	1·392	20 21	1·056	116 48	1·302	210 30
	45	1·414	·577	·471	·962	1·414	·577	1·527	22 12	1·071	116 6	1·528	202 30
Parab.	50	1·556	·688	·487	·974	1·925	·440	1·701	23 51	1·088	116 35	1·975	192 30
	51° 0′	1·589	·713	·491	·976	2·077	·400	1·741	24 9	1·093	116 43	2·115	190 30
	52	1·624	·739	·495	·978	2·256	·353	1·787	24 28	1·097	116 51	2·283	188 30
	54	1·701	·795	·504	·984	2·729	·229	1·878	25 2	1·105	117 7	2·738	184 30
Hyperbs.	55	1·743	·824	·509	·986	3·049	·145	1·928	25 18	1·109	117 16	3·053	182 30
	56° 18′	1·802	·866	·515	·990	3·601	·000	1·999	25 39	1·116	117 30	3·601	180 30
	59	1·942	·961	·530	·997	5·786	+·566	2·166	26 20	1·129	118 59	5·813	174 30
Line	60	2·000	1·000	·536	1·000	7·468	1·000	2·236	26 34	1·134	118 11	7·534	172 30
	61	2·063	1·042	·542	1·003	−10·53	+1·793	2·311	26 48	1·140	118 24	10·68	170 30
Convex {	63° 26′ − ε	2·236	1·155	·559	1·010	∞	∞	2·517	27 19	1·155	118 57	∞	165 30
	63° 26′ + ε					−∞	∞					−∞	134 30
	64	2·281	1·184	·563	1·012	+45·22	−12·60	2·570	27 26	1·157	119 6	46·94	344 30
Hyperbs.{	65	2·366	1·238	·571	1·015	16·36	5·146	2·670	27 37	1·165	119 21	17·15	342 30
Parab.	65° 52′	2·446	1·289	·578	1·019	10·12	3·552	2·765	27 47	1·171	119 35	10·80	340 30
	66	2·459	1·297	·579	1·019	9·987	3·500	2·779	27 48	1·172	119 37	10·59	340 30
	68	2·669	1·429	·596	1·026	5·617	2·369	3·028	28 10	1·186	120 11	6·090	337 30
	70	2·924	1·586	·616	1·033	3·912	1·927	3·326	28 29	1·202	120 48	4·360	335 30
Ellipses	72	3·237	1·777	·638	1·041	3·008	1·694	3·693	28 47	1·221	121 29	3·455	330 30
	75	3·864	2·155	·674	1·054	2·230	1·488	4·424	29 9	1·251	122 36	2·681	325 30
	80	5·759	3·274	·751	1·079	1·568	1·312	6·624	29 37	1·315	124 49	2·045	320 30
	85	11·47	6·599	·854	1·112	1·217	1·216	13·25	29 54	1·402	127 32	1·720	315 30
Parab.	89° 20′	86·41	49·79	·979	1·148	1·024	1·161	99·5	29 56	1·508	130 24	1·548	311 30
Hyperbs.	90 − ε	∞	∞	1·000	1·155	+1·000	−1·155	∞	30	1·527	130 54	1·527	310 30

c	Eq. of Orbit	e	ϖ	a	γ_{12}	γ_{23}	γ_{31}	T_{12}	T_{23}	T_{31}
0°	·000 ·000 + 1·000	·000	ind.°	1·000	1·732	1·732	1·732	1·000	1·000	1·000
5	+·0101 − ·0015 1·0104	·010	171 19	1·010	1·729	1·832	1·678	·987	1·106	·953
10	·039 ·014 1·046	·041	160 52	1·048	1·724	1·991	1·668	·956	1·316	·946
15										
20	·083 ·061 1·126	·103	143 48	1·138	1·718	2·244	1·710	·924	1·777	·962
25										
30	·135 ·209 1·317	·248	122 54	1·404	1·740	2·684	1·826	·878	3·238	·966
35										
40	·161 ·395 1·527	·426	112 12	1·867	1·805	3·055	1·925			
45	·186 ·815 1·972	·836	102 50	6·554	2·016	3·659	2·063	·878	48·60	·849
50	·191 ·982 2·140	1·000	101 14	∞	2·100	3·831	2·097	·879	∞	·820
51° 0′	·196 1·150 2·319	1·166	99 39	6434						
52										
54	·203 1·719 2·898	1·720	96 44	1·481						
55	·207 2·187 3·366	2·192	95 25	·855	2·781	4·890	2·258	·895	∼	·665
56° 18′	·212 3·074 4·227	3·081	93 56	·498						
59	·221 − 1·415 + 15·42	1·415	90 30	·077						
60	·224 ∞ ∞$(y-1)$	∞	90	·000	6·932	9·468	2·536	·000	∼	·000
61					Convex Orbits.					
63° 26′ − ε	·234 + 4·906 − 3·671	4·912	87 17	·159						
63° 26′ + ε	·578 − 1·761 + 3·257	1·853	108 11	1·338	∞	∞	2·799	∼	∞	·909
64										
65	·587 ·979 2·134	2·494	120 21	8·666	18·014	15·380	2·945			
65° 52′	·591 ·805 2·257	1·000	126 19	∞	11·633	9·072	3·036	∞	7·746	1·386
66	·593 ·779 2·221	·979	127 15	5·383						
68	·606 ·338 1·894	·693	150 53	3·645						
70	·619 − ·120 1·708	·630	169	2·834	5·409	3·649	3·584	5·735	6·343	2·685
72	·635 + ·027 1·599	·636	182 25	2·674						
75	·654 ·185 1·497	·680	195 47	2·783	3·859	3·981	4·644	2·62	6·68	4·64
80	·692 ·366 1·439	·783	207 52	3·716	3·327	6·212	6·870	1·97	10·21	9·343
85	·740 ·514 1·455	·892	214 47	7·721	3·115	12·895	13·480			
89° 20′	·764 ·645 1·505	1·000	219 51	∞	3·055	9·943	102·7	1·200	225·4	∞
90 − ε	·770 + ·666 + 1·527	1·018	220 6	41·000	3·055	∞	∞	1·148	∞	∼

[p. 452]

Article Nos. 95 *to* 98. *Planogram No.* 3, *the Orbit-pole at one of the points A.*

95. When the orbit-pole is at one of the points A, the orbit-plane passes through one of the rays, and as there is no longer on this ray any determinate point of intersection, the orbit (as was seen) becomes indeterminate. Thus consider the point A for which $b = 270°$, $c = 60°$: we have

$$\alpha,\ \beta,\ \gamma = -1,\ 0,\ 0,$$
$$\alpha',\ \beta',\ \gamma' = 0,\ -\tfrac{1}{2},\ -\tfrac{1}{2}\sqrt{3},$$
$$\alpha'',\ \beta'',\ \gamma'' = 0,\ -\tfrac{1}{2}\sqrt{3},\ \tfrac{1}{2},$$

[Figure 9: a circle in the (x, y)*-plane of unit radius, with the four cardinal positions of points* 1, 2, 3, B *marked. Point* 1 *at the top,* 2 *at lower left,* 3 *at lower right,* B *at the right.]*

and consequently the formula gives

$$x_1' : y_1' : 1 = 0 : 0 : 0,$$
$$x_2' : y_2' : 1 = -\tfrac{1}{2}\sqrt{3} - \sqrt{3} : 3 : -\tfrac{1}{2}\sqrt{3} - \sqrt{3},$$
$$x_3' : y_3' : 1 = -\tfrac{1}{2}\sqrt{3} - \sqrt{3} : -3 : -\tfrac{1}{2}\sqrt{3} - \sqrt{3},$$

and, moreover, $x = -x'$, $y = -y'$. From the formula the value of x_1' or x_1 is given as $\frac{0}{0}$, but the true value is obviously $x_1 = 1$; the value of y_1 is actually indeterminate. The formulae give the values of (x_2, y_2), (x_3, y_3), viz. the system is

$$x_1 = 1, \quad y_1 = \text{ind.}$$
$$x_2 = -1, \quad y_2 = \tfrac{2}{\sqrt{3}}, \qquad \text{whence } r_2 = r_3 = \sqrt{\tfrac{7}{3}},$$
$$x_3 = -1, \quad y_3 = -\tfrac{2}{\sqrt{3}},$$

[p. 453]

so that the orbits in the planogram are the whole series of conics having a given focus, S, and passing through two fixed points, 2, 3, having the common abscissa $x = -1$, and at equal distances $\frac{2}{\sqrt{3}}$ $(= 1{\cdot}15470)$ on opposite sides of the axis. The axis of x is obviously the common transverse axis for all the orbits; that is, the equation of the orbit will be of the form $r = Ax + B$; and writing $x = -1$, we have $\sqrt{\frac{7}{3}} = -A + B$, viz. the equation is $r - \sqrt{\frac{7}{3}} = A(x + 1)$; the value of A will be determined if we assume for the point 1 a determinate position on the line $x = 1$, say its ordinate is $= y_1$; for then if $r_1 = \sqrt{1 + y_1^2}$ we have $r_1 - \sqrt{\frac{7}{3}} = 2A$, and the equation is $r - \sqrt{\frac{7}{3}} = \frac{1}{2}(r_1 - \sqrt{\frac{7}{3}})(x + 1)$. In particular if $y_1 = 0$, we have $r_1 = 1$, and the equation of the orbit is $r - \sqrt{\frac{7}{3}} = \frac{1}{2}(1 - \sqrt{\frac{7}{3}})(x + 1)$: this is the orbit, eccentricity $= \frac{1}{2}(\sqrt{\frac{7}{3}} - 1) = {\cdot}264$, belonging to the point A as a point in planogram No. 1: for the value of y_1 being in that planogram originally assumed $= 0$, is of course $= 0$ when the orbit-pole comes to be the point A.

96. We may conversely take the equation of the orbit, or say the value of $A\,(= \pm e)$ in the equation $r - \sqrt{\frac{7}{3}} = A(x + 1)$, to be given; and then writing $x = x_1 = 1$, we have

$$r_1 = \sqrt{\tfrac{7}{3}} + 2A, \quad \text{that is} \quad y_1^2 = (\sqrt{\tfrac{7}{3}} + 2A)^2 - 1;$$

for

$$r_1 = 1 \quad \text{or} \quad y_1 = 0, \quad A = \tfrac{1}{2}(1 - \sqrt{\tfrac{7}{3}}) = -{\cdot}264,$$

and as r_1 increases to $r_1 = \sqrt{\frac{7}{3}}$, or y_1 increases to $+\frac{2}{\sqrt{3}}$, A diminishes from $-{\cdot}264$ to 0; viz., for $r_1 = \sqrt{\frac{7}{3}}$ or $y_1 = \frac{2}{\sqrt{3}}$, the orbit is a circle; as r_1 increases from $\sqrt{\frac{7}{3}}$, or y_1 from $+\frac{2}{\sqrt{3}}$, A increases from 0 positively;

for $r_1 = \sqrt{\frac{7}{3}} + 2 = 3{\cdot}527$, or $y_1 = +\frac{1}{\sqrt{3}}\sqrt{\ } = 2{\cdot}896$, A becomes $= 1$; that is, the orbit is a parabola; and for larger positive values of r_1, or positive or negative values of y_1, the orbit is a hyperbola (concave); and ultimately for $r_1 = \infty$ or $y_1 = \pm\infty$, the orbit is the right line $x + 1 = 0$. Thus A extends from $-{\cdot}264$ to 0, and thence from 0 positively to ∞.

97. In further illustration, suppose that the orbit-pole, instead of being at A, is a point in the immediate neighbourhood of A, say that the rectangular spherical coordinates, measured from A in the direction of the meridian and perpendicular thereto, are ξ and η, the colatitude and longitude of the orbit-pole being thus $c = 60° + \xi$, and $b = 270° + \dfrac{2}{\sqrt{3}}\eta$; we have then, ξ, η being indefinitely small,

$$\begin{aligned}
\alpha,\ \beta,\ \gamma &= -1,\ -\tfrac{2}{\sqrt{3}}\eta,\ 0,\\
\alpha',\ \beta',\ \gamma' &= \tfrac{2}{\sqrt{3}}\eta\,,\ -\tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\xi\,,\ -\tfrac{\sqrt{3}}{2} + \tfrac{1}{2}\xi\,,\\
\alpha'',\ \beta'',\ \gamma'' &= \eta\,,\ -\tfrac{\sqrt{3}}{2} + \tfrac{1}{2}\xi\,,\ \tfrac{1}{2} - \tfrac{\sqrt{3}}{2}\xi\,.
\end{aligned}$$